

FLUXION CANDIDATE ASSESSMENT

Strategic Problem-Solving Case Study

A note before you begin

Please take a few minutes to read this document in full before writing a single word. Reviewing everything beforehand ensures you don't miss any critical details or begin the task only to realize there was more context required.

What you are about to begin working on is not a test in the traditional sense. There are no answer keys or model solutions. In fact, there is not even an expectation that you will be able to resolve everything that is put in front of you. That is because we are evaluating your process, not necessarily your conclusion.

We have given you four days for this because we want you to really 'wrestle' with the problem. Take the time to look at it, walk away, and come back with fresh eyes. Please note that if a submission is clearly written in a single afternoon, the evaluation will reflect that lack of depth

Furthermore, we want to emphasize that we are not assessing whether you can solve the problems of every discipline simultaneously. We are assessing whether you can solve the problem from where you stand. A mechanical engineer should think like a mechanical engineer; a data scientist should think like a data scientist. Each discipline has a specific lens through which it views the world, and each perspective is valuable. Your focus area is not a limitation; instead, it provides a unique vantage point that we want to explore.

Throughout your responses, you are expected to be explicit about how your background is shaping your reasoning. When you choose an approach, tell us why someone from your discipline would choose that approach. When you deprioritize something, tell us whether that is a principled analytical choice or simply outside your domain. We are not looking for someone who can do everything. We are looking for someone who understands deeply what they can contribute and where the boundary of that contribution lies. The final destination of any real project is reached by a team. Your job is to show us how you would contribute to that team from your specific position, not to demonstrate that you could replace the entire team alone.

Section 1 About you

1.1

State the role you are applying for and the discipline or background you are coming from. One sentence is enough.

1.2

Describe a specific experience, either from your studies, a project, an internship, or anywhere else, that made you genuinely interested in the work you do. We are not looking for motivation-letter language. We are looking for a specific moment or realization, described honestly. If it was a textbook problem that clicked unexpectedly, say that. If it was watching something fail and wanting to understand why, say that.

This response has a strict limit of **150 words**. Responses that exceed this limit will not be reviewed. **The limit is intentional**. We want to see whether you can say something real without adding all the unnecessary fluff around it..

Section 2 The Scenario

Narayan Industrial Township (NIT)

Narayan Industrial Township is a 1,200-acre integrated industrial zone commissioned four years ago in a semi-arid coastal region of a developing economy. It was conceived as a coordinated system, manufacturing, logistics, worker housing, healthcare, utilities, and shared data infrastructure all built together, with the expectation that each component would reinforce the others. Stable power would enable manufacturing. Manufacturing revenues would fund housing and healthcare. Housed and healthy workers would reduce attrition. Reliable logistics would sustain supply and dispatch. A common data backbone would give every operator visibility across the whole.

That is not how it has unfolded.

NIT is currently operating at 34% of its designed capacity. Over the last twelve months, it has been generating ₹11 crore per month in revenues against ₹18 crore per month in operating costs. The governing authority, which built NIT through a combination of public and private co-investment, has brought your team in as consultants. They have allocated ₹45 crore and given you 18 months to materially reverse its trajectory.

The following that is below covers the operational data from the last 12 months.

Operational Data

Metric	Designed	Actual
Manufacturing units active	12	5
Logistics hub throughput (trucks/days)	600	110
Medical center occupancy (% of designed capacity)	100%	340% during shift changes
Average power uptime across zones	99.2%	71%
Workforce annual attrition	~18% (industry norm)	67%
Average equipment downtime per active manufacturing unit	<5%	34%
Water supply failure events (12 months)	0-2	23
Data and telecom connectivity in Zone D	4G minimum	3G in 2 of 4 sub-zones
Manufacturing units that attempted operations and subsequently shut down	—	3
Manufacturing units that never launched	—	4

Facility Zones

Zone A — Light manufacturing. 12 units designed. 5 currently operational. 3 attempted operations and have since shut down. 4 have never launched.

Zone B — Logistics and warehousing. 3 warehouses designed. 1 operational. Two logistics companies are contracted to operate this zone; one has formally communicated intent to exit.

Zone C — Worker housing and healthcare. Approximately 4,200 residents. A 120-bed medical center serves the township. The nearest alternative facility is a regional hospital 38 km away, which is currently absorbing overflow.

Zone D — Shared infrastructure. Houses the data center, power substation, water treatment plant, and telecom exchange. This is the backbone zone.

Zone E — Administrative and commercial. Offices, vendor units, banking, and staff canteen.

Stakeholders on Record

The governing authority, which holds ultimate accountability for NIT's performance. Fourteen industrial operators, some active, some stalled, three who have departed. Approximately 4,200 workers and their families in Zone C. The regional hospital absorbing medical overflow. Two logistics operators in Zone B. A telecom provider currently in a contractual dispute with the township authority over the Zone D SLA.

What You Are Being Asked To Do

You are not being asked to fix NIT completely. You are being asked to think clearly and honestly about what should be fixed first, in what sequence, with what resources, and, just as importantly, what you are choosing not to address and why. An answer that claims to solve everything is not more impressive. It is less credible.

Section 3 Problem Framing

3.1

Before forming any hypothesis, identify the eight questions you would most need answered to understand what is actually happening at NIT. For each question, write **one sentence** explaining why that question is load-bearing, meaning, how your thinking changes depending on the answer. Do not list questions that are merely interesting. Only include questions whose answers would materially alter your approach.

3.2

Based only on the data provided above, without any assumptions beyond what is written, state your single best hypothesis about the primary cause of NIT's underperformance. State it in **two sentences**. Then state your confidence level (low, medium, or high) and explain in **two further sentences** why you are not more confident than that.

3.3

The data table contains at least two patterns that a careful analyst should find suspicious. Identify them. You do not need to explain them, just name them precisely and state what makes each one worth questioning.

Section 4 System Decomposition

4.1

Describe NIT as a system. Not as a list of zones since that information has already been given to you. We are asking you to identify the structural dependencies: what depends on what, what flows through what, where failures propagate and why. Use whatever format your discipline uses for this kind of analysis, like a dependency map described in text, or an input-output model, or even a failure propagation framework. The format matters less than the quality of what you are seeing.

4.2

In a system with multiple simultaneous failures, there is typically a binding constraint, like one failure whose resolution would unlock the most recovery across the rest of the system. Identify what you believe is NIT's binding constraint and make the case for it. Be clear about the distinction you are drawing: a binding constraint is not simply the largest problem. It is the problem that is blocking the resolution of most other problems.

4.3

The 67% attrition figure appears in the data table alongside power uptime and equipment downtime, as though it is a metric of equivalent type. It is not. Make the argument, with structure, for whether workforce attrition at NIT is a cause of the township's failures, an effect of them, or both and then trace the feedback loop if one exists.

Section 5 Diagnosis

5.1

The medical center is operating at 340% of its designed capacity during shift changes. The design capacity is 120 beds. Construct three distinct hypotheses for why this is occurring. For each hypothesis, describe one specific piece of data you would collect to confirm or rule it out.

5.2

Three manufacturing units attempted operations and shut down. Four never launched. These are meaningfully different failure modes. From a systemic perspective, what do you believe most likely distinguishes them? What does this distinction, if it is a real one, suggest about the nature of NIT's failure more broadly?

5.3

Power uptime stands at 71% against a designed 99.2%. Trace the second and third-order consequences of this single failure across at least four other metrics in the dataset. Show the propagation explicitly and do not summarize it. We want to see the chain.

Section 6 Intervention Design

This section is the center of your submission.

You have ₹45 crore and 18 months.

6.1 — How you decided what to prioritize

Before naming a single intervention, explain the logic by which you arrived at your priorities. What framework, model, or analytical method did you use? This should be written explicitly enough that someone else could apply the same logic to a different problem. Intuition is not a framework.

6.2 — What you would actually do

Describe your interventions specifically. General recommendations like whether to improve infrastructure, invest in workforce development, and upgrade connectivity, are not acceptable responses here. Every intervention you propose must include: a defined scope, a rationale tied directly to your earlier diagnosis, a defensible allocation of resources, and at least one measurable output that would tell you six months later whether the intervention is working.

6.3 — Sequencing

In what order do your interventions occur, and why does that order matter? A plan with no sequencing logic is a list of good intentions. Show the dependencies in your intervention design the same way you showed the dependencies in the system itself.

6.4 — What you are not doing

Identify at least two things present in the data that you are consciously choosing not to address within this intervention period. Explain why. Deprioritization is a decision, and it should be as defensible as anything you chose to include.

6.5 — Discipline-specific depth

This is where we evaluate your actual technical competence, not your ability to think at the strategic level. Go deep on the component of this problem that falls most directly within your own domain and discipline. For example, a data scientist should demonstrate rigor in data,

modeling, or systems analysis. A mechanical engineer should demonstrate rigor in physical systems, equipment, or failure analysis. Do not remain at the strategic level in this section. Show your craft.

As you write this section, make clear how your disciplinary background is shaping the choices you are making. When you choose an approach, say why someone with your training would approach it that way. This is not a formality but rather it is central to how we evaluate whether you understand your own contribution.

Section 7 Testing Your Own Plan

7.1

Identify the single assumption in your intervention plan that, if incorrect, would most significantly undermine what you have proposed. What is the assumption? What early signal would tell you it is wrong? What would you do if you received that signal at month four?

7.2

A colleague reviews your plan and raises the following concern: **"You have optimized for recovery speed, but in doing so you have created a new single point of failure."**

Without knowing the specifics of your plan, describe how this critique might apply to it and state honestly whether you think the critique is valid.

7.3

You are six months into the plan. Two interventions are on track. One has stalled due to a contractor failure. The governing authority asks whether to continue or cut losses on the delayed workstream. Walk through the decision. What information do you need before answering? What is the decision framework?

7.4

Three months before the end of the 18-month period, you are informed that the budget has been reduced from ₹45 crore to ₹28 crore, effective immediately. What changes in your plan? What does not change? Explain both.

Section 8 Open Response

NIT was not the product of careless design; the engineers behind it were undeniably competent. Its power infrastructure met every appropriate benchmark, and the medical center was sized according to the standard formulas for a facility of its scale. From a logistics hub that aligned with industry standards to data specifications that held up under scrutiny, the technical foundation was, on the surface, exactly what it needed to be.

And yet the township is failing.

In **300 to 500 words**, respond to the following:

What does NIT tell you about the difference between designing a system and designing a system that works?

Use NIT as your primary evidence. You may draw on any idea, field, or experience that is genuinely relevant to your argument, but do not use borrowed frameworks as a substitute for thinking. We are not looking for an essay on complexity theory. We are looking for a specific argument, grounded in what you have actually observed in this case.

This is the section we will spend the most time reading.

How We Will Evaluate Your Submission

We are sharing this so there is no ambiguity about what we are looking for.

What we are looking at	What a weak response looks like	What a strong response looks like
Problem framing	Generic questions, premature solutions	Questions that reveal hidden assumptions in the scenario
System decomposition	Restating the zone list	Identifying real structural dependencies
Binding constraint	Treating all failures as equally weighted	Naming one constraint and defending it rigorously
Intervention design	Vague, unsequenced, no trade-offs acknowledged	Specific, ordered, honest about what was sacrificed
Discipline-specific depth	Staying at the strategic level throughout	Demonstrating actual craft within your domain
Disciplinary self-awareness	Trying to cover every angle equally	Being clear about where your contribution lies and where it ends
Intellectual honesty	Overconfidence, no acknowledgment of limits	Knows what it does not know and says so precisely
Communication	Padded, jargon-heavy, circular	Dense, specific, no sentence that earns its place by sounding serious

We look forward to reading your submission.